INCH-POUND

MIL-T-27/223C
4 May 2012
SUPERSEDING
MIL-T-27/223B
30 June 1992

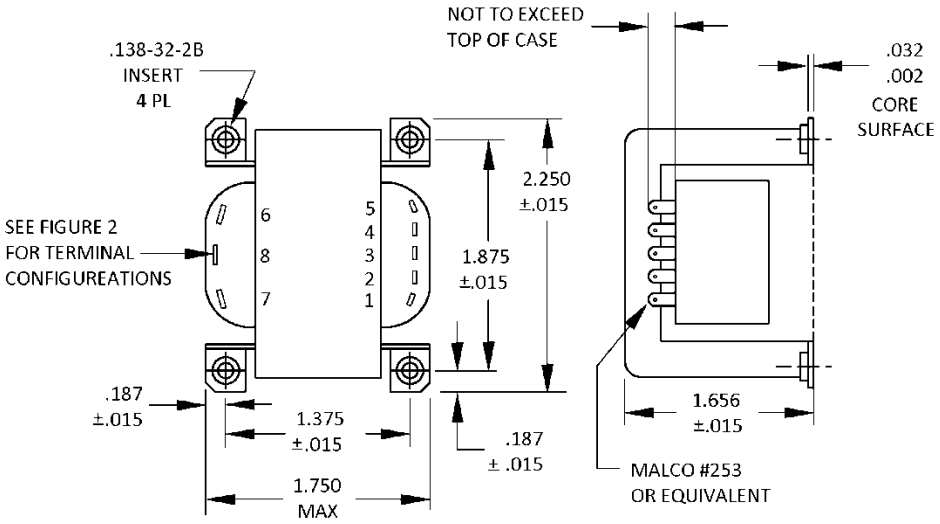
MILITARY SPECIFICATION SHEET

TRANSFORMERS, POWER, STEP-DOWN

Inactive for new design after 30 June 1992.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The number of terminals and the configuration of the terminals shall be as shown on figure 2.
4. For circuit diagram and marking, see figure 3.
5. Marking shall be on side of case.

FIGURE 1. Dimensions and configuration.

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Inches	mm
.002	0.05
.015	0.38
.032	0.81
.138	3.51
.187	4.75
1.375	34.92
1.656	42.06
1.750	44.45
1.875	47.62
2.250	57.15

FIGURE 1. Dimensions and configurations - continued

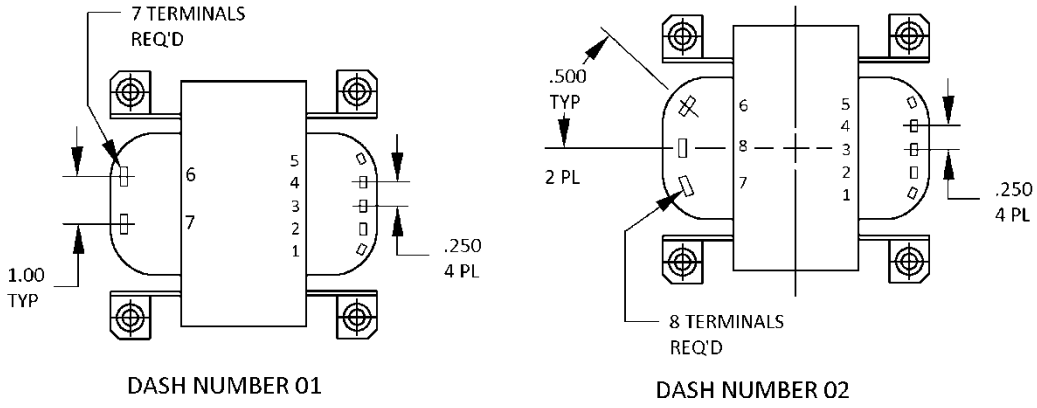
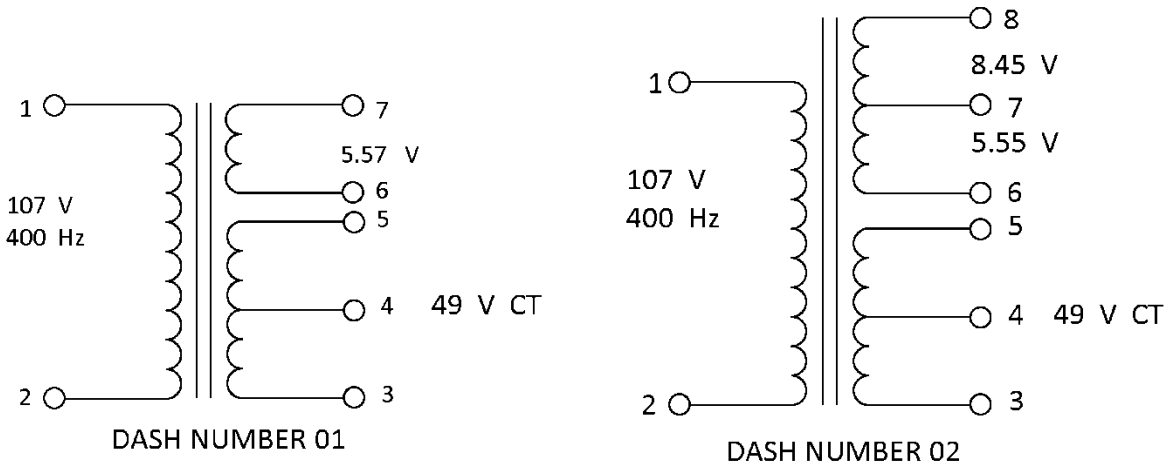


Figure 2. Terminal configuration

Inches	mm
.250	6.35
.500	12.70
1.000	25.40

NOTES:
1. Dimensions are in inches.
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NOTE: Working voltage = 120 v rms, altitude = 70,000 feet maximum

Figure 3. Circuit diagram and marking

REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings: See table I.

Primary voltage (1-2): 107 V rms, 400 Hz ±20.

Working voltage: 120 V rms.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Terminals: Solder terminal Malco no. 253 or equivalent.

Terminal height: Not to exceed top of case.

Weight: 11.8 ounces maximum.

Altitude: 70,000 feet maximum.

Operating temperature range: -55°C to +170°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds.

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Dielectric withstanding voltage:

At sea level: 1,000 V rms.

At reduced barometric pressure: 300 V rms.

Electrical characteristics:

Rated load: With 107 V rms, 400 Hz across (1-2), the secondary voltage shall be as specified (see table I).

Polarity: Polarity shall be the same on terminals (see table II).

Temperature rise: 99°C with 107 V rms, 400 Hz across (1-2) at an ambient temperature of 71°C.

VERIFICATION:

Qualification inspection: Not applicable.

Conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/223-(dash number from table I).

TABLE I. Electrical ratings.

Dash no.	Volt ampere rating (VA)	Voltage (V rms ±10%)			Current (A)			
		(3-5)	(6-7)	(7-8)	(1-2)	(3-5)	(6-7)	(7-8)
01	125	49 CT	5.57	---	1.20	0.010	22.5	---
02	130	49 CT	5.55	8.45	1.25	0.010	16.4	2.86

TABLE II. Electrical characteristics.

Dash no.	No load: With 107 V rms, 400 Hz in (1-2)					DC resistance (Ω max)				Polarity
	Current mA, (max) (1-2)	Power W (max) (1-2)	Voltage ±10% (3-5)	Voltage ±10% (6-7)	Voltage ±10% (7-8)	(1-2)	(3-5)	(6-7)	(7-8)	
01	350	42	49 CT	5.57	---	1.93	20.0	.007	---	1, 3, 6
02	350	42	49 CT	5.55	8.45	1.93	20.0	0.14	0.1	1, 3, 6

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references no other documents.

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5950-2009-081

Review Activities:
Army – AR, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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